

General Organic Chemistry (GOC)

JEE Chemistry — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Organic compounds are classified by their functional group - the reactive part of the molecule that determines its chemical behavior (e.g., -OH for alcohols, -COOH for carboxylic acids).
- Inductive effect (I effect) is the permanent displacement of sigma-bond electrons along a chain due to electronegativity difference; -I groups withdraw electron density, +I groups donate it.
- Resonance (mesomeric effect) involves delocalization of pi electrons or lone pairs across a conjugated system, stabilizing the molecule and affecting reactivity.
- Hyperconjugation is the interaction of C-H sigma bonds adjacent to a positive center (or double bond) with the pi system, stabilizing carbocations and alkenes.
- Electrophiles are electron-deficient species that accept electron pairs (e.g., H^+ , NO_2^+); nucleophiles are electron-rich species that donate electron pairs (e.g., OH^- , CN^-).
- Types of organic reactions: substitution (one group replaces another), addition (atoms add across a multiple bond), elimination (atoms are removed to form a multiple bond), and rearrangement.
- Isomers are compounds with the same molecular formula but different structures; types include chain, position, functional group isomerism (structural) and geometrical, optical isomerism (stereoisomerism).



Carbon skeleton with a functional group

A carbon skeleton with a functional group (-OH) and an inductive effect shown along the chain.

2. Formula Sheet

General formula, alkanes	$C_n H_{(2n+2)}$
General formula, alkenes	$C_n H_{2n}$
General formula, alkynes	$C_n H_{(2n-2)}$
Degree of unsaturation	$DoU = (2C + 2 + N - H) / 2$
Inductive effect order (electron withdrawing)	$-NO_2 > -CN > -F > -Cl > -Br > -I$ (approx)
Stability of carbocations	3 degree > 2 degree > 1 degree > methyl
Nucleophilicity (general trend in same period)	Increases with electron density/basicity
Optical isomer condition	Chiral carbon: 4 different groups attached

3. Practice Problems (22 questions)

1. Identify the functional group present in CH_3-CH_2-OH .
2. Find the degree of unsaturation for the molecular formula C_6H_6 (benzene).
3. State whether $-NO_2$ is an electron-withdrawing (-I) or electron-donating (+I) group.
4. Explain the difference between the inductive effect and the resonance (mesomeric) effect.
5. Rank the following carbocations in order of increasing stability: methyl, primary, secondary, tertiary.
6. Define a nucleophile and give two examples.
7. State the type of isomerism shown by butane and isobutane (2-methylpropane).
8. Explain what is meant by a chiral carbon and state the condition for a carbon atom to be chiral.
9. Classify the following reaction type: $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ (in the presence of light).
10. Explain hyperconjugation and state its effect on the stability of alkenes and carbocations.
11. Find the general molecular formula for an alkyne with n carbon atoms.
12. State the difference between homolytic and heterolytic bond cleavage.
13. Explain why tertiary carbocations are more stable than primary carbocations, using the inductive and hyperconjugation effects.
14. Identify the type of isomerism between propan-1-ol and propan-2-ol.
15. Define electrophile and give two examples of common electrophiles in organic reactions.
16. State the IUPAC rule for numbering the longest carbon chain in an organic compound (lowest locant rule, briefly).
17. Explain why alcohols have higher boiling points than alkanes of similar molecular mass.

18. Classify the following as substitution, addition, or elimination: $\text{CH}_2=\text{CH}_2 + \text{H}_2 \rightarrow \text{CH}_3-\text{CH}_3$.
19. State the meaning of 'geometrical isomerism' and the structural feature required for it to occur (e.g., restricted rotation around a double bond).
20. Find the number of sigma and pi bonds in an alkyne triple bond (C tripe bond C).
21. Explain the difference between structural isomerism and stereoisomerism, with one example of each.
22. State why fluorine, despite being highly electronegative, shows a +M (electron donating resonance) effect in some contexts alongside its -I effect.

Answer Key

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| 1. -OH (hydroxyl group, alcohol) | 2. DoU = 4 (3 double bonds + 1 ring, consistent with benzene) |
| 3. Electron-withdrawing (-I effect) | 4. Conceptual - inductive effect acts through sigma bonds (short range, permanent); resonance involves pi electron/lone pair delocalization (through conjugated pi systems) |
| 5. Methyl < primary < secondary < tertiary (increasing stability) | 6. Conceptual - electron-rich species that donates an electron pair; e.g., OH ⁻ , CN ⁻ |
| 7. Chain isomerism (same functional group, different carbon skeleton) | 8. Conceptual - a carbon atom bonded to four different groups/atoms, making the molecule optically active |
| 9. Substitution reaction (free radical substitution) | 10. Conceptual - adjacent C-H sigma bonds align with an empty p-orbital or pi system, donating electron density and providing extra stability |
| 11. $\text{C}_n\text{H}_{(2n-2)}$ | 12. Conceptual - homolytic cleavage gives two radicals (each atom keeps one electron); heterolytic cleavage gives two ions (one atom keeps both electrons) |
| 13. Conceptual - more alkyl groups around the positive carbon provide more +I donation and more hyperconjugation, stabilizing the tertiary carbocation | 14. Position isomerism (same functional group -OH, different position on the chain) |
| 15. Conceptual - electron-deficient species that accepts an electron pair; e.g., H ⁺ , NO ₂ ⁺ | 16. Conceptual - longest chain is numbered to give the lowest possible locants to substituents/functional groups |
| 17. Conceptual - alcohols can form hydrogen bonds (O-H...O) between molecules, requiring more energy to separate, raising boiling point | 18. Addition reaction (hydrogenation) |
| 19. Conceptual - occurs when a carbon-carbon double bond has restricted rotation and each doubly-bonded carbon has two different substituents, giving cis/trans (or E/Z) forms | 20. 1 sigma + 2 pi bonds |
| 21. Conceptual - structural isomers have different connectivity (e.g., butane vs isobutane); stereoisomers have the same connectivity but different spatial arrangement (e.g., cis/trans-2-butene) | 22. Conceptual - fluorine's lone pairs can be donated into an adjacent pi system (resonance donation, +M), even though its high electronegativity gives a strong -I effect through sigma bonds |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.